DEPARTMENT OF Electrical Engineering
UNIVERSITY OF ENGINEERING & TECHNOLOGY, PESHAWAR
JALOZAI CAMPUS

Course: Computer Programming (2nd Semester)
Assignment: Assignment No. 1 (Spring-2026)
Student Name: Farhan Amin
Registration No: BF25NWELE0745

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i. Variable Declaration :

```
#include <iostream>
using namespace std;

int main() {
    // Defining these as double since electrical values often have decimals
    double voltage, current, resistance;
    return 0;
}
```

ii. Display Message :

```
#include <iostream>
using namespace std;

int main() {
    // Just a simple print statement to welcome the user
    cout << "Welcome to Object Oriented Programming in C++!";
    return 0;
}
```

iii. Read Age :

```
#include <iostream>
using namespace std;

int main() {
    int age;
    cout << "Please enter your age: ";
    cin >> age; // Saving the user input into the age variable
    return 0;
}
```

iv. Temperature Check :

```
#include <iostream>
using namespace std;

int main() {
```

```
    int temperature = 105; // Testing with a value over 100
    if (temperature > 100) {
        cout << "Overheating detected!"; // This only runs if it's too hot
    }
    return 0;
}
```

v. Area of a Circle :

```
#include <iostream>
using namespace std;

int main() {
    double r = 5.0, area;
    // Using the constant PI value given in the assignment
    area = 3.14159 * r * r;
    cout << "The area of the circle is: " << area;
    return 0;
}
```

vi. Print x, y, and z :

```
#include <iostream>
using namespace std;

int main() {
    int x = 10, y = 20, z = 30;
    // Printing them all out with commas in between
    cout << x << ", " << y << ", " << z;
    return 0;
}
```

vii. Sum of Voltage and Current :

```
#include <iostream>
using namespace std;

int main() {
    double voltage = 12.0, current = 0.5, total;
    total = voltage + current; // Adding them together
    return 0;
}
```

viii. Division by Zero Guard :

```
#include <iostream>
using namespace std;

int main() {
    int denominator = 0;
    // Safety check to make sure we don't divide by zero
    if (denominator == 0) {
        cout << "Error: Division by zero";
    }
    return 0;
}
```

ix. Increment and Print :

```
#include <iostream>
using namespace std;

int main() {
    int count = 10;
    // Incrementing first then printing the new value
    cout << ++count;
    return 0;
}
```

x. Read Three Floats :

```
#include <iostream>
using namespace std;

int main() {
    float a, b, c;
    cout << "Enter three floating-point numbers: ";
    cin >> a >> b >> c; // Getting all three at once
    return 0;
}
```

xi. Total Resistance (Series) :

```
#include <iostream>
using namespace std;

int main() {
    double r1, r2, r3, total;
    cout << "Enter values for 3 series resistors: ";
    cin >> r1 >> r2 >> r3;
    total = r1 + r2 + r3; // Series is just the sum
    cout << "Total Series Resistance: " << total;
    return 0;
}
```

xii. Total Resistance (Parallel) :

```
#include <iostream>
using namespace std;

int main() {
    double r1, r2, r3, total;
    cout << "Enter values for 3 parallel resistors: ";
    cin >> r1 >> r2 >> r3;
    // Using the reciprocal formula for parallel circuits
    total = 1.0 / (1.0/r1 + 1.0/r2 + 1.0/r3);
    cout << "Total Parallel Resistance: " << total;
    return 0;
}
```

xiii. Voltage Division Rule (VDR) :

```
#include <iostream>
using namespace std;
```

```
int main() {
    // Picking some standard values for the circuit
    double R1 = 50, R2 = 100, V_source = 15, VR2;
    // VDR formula: (Source Voltage * R2) / Total Resistance
    VR2 = (V_source * R2) / (R1 + R2);
    cout << "The voltage across R2 is: " << VR2;
    return 0;
}
```

xiv. Current Division Rule (CDR) :

```
#include <iostream>
using namespace std;

int main() {
    // Selecting resistors and total current
    double R1 = 10, R2 = 40, I_total = 2.5, IR2;
    // CDR formula: (Total Current * R1) / (R1 + R2)
    IR2 = (I_total * R1) / (R1 + R2);
    cout << "The current through R2 is: " << IR2;
    return 0;
}
```

xv. Semester GPA Calculation :

```
#include <iostream>
using namespace std;

int main() {
    double g1, g2, g3, c1, c2, c3, gpa;
    cout << "Enter GPA and credits for 3 subjects: ";
    cin >> g1 >> c1 >> g2 >> c2 >> g3 >> c3;
    // Multiplying each grade by its credits and dividing by total credits
    gpa = (g1*c1 + g2*c2 + g3*c3) / (c1 + c2 + c3);
    cout << "My semester GPA is: " << gpa;
    return 0;
}
```